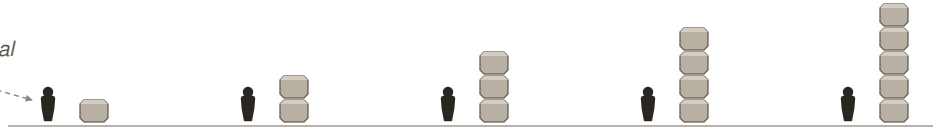


LAYER 1 Within-group signaling

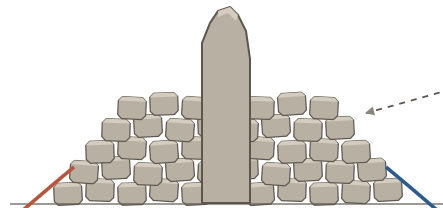
Each construction act is a costly, publicly witnessed contribution:
a credibility-enhancing display (CRED).

Contribution scales with capacity (Spence separating equilibrium)

*private capacity q
hidden; figures identical*



construction acts accumulate



Monument M_g

accumulated evidence of collective construction

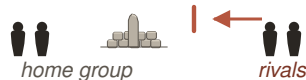
*peers see it,
positional reward β_o*

rivals read it

partners read it

LAYER 2 Competition

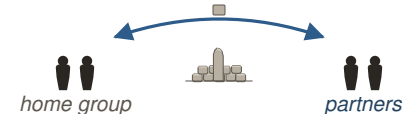
A rival assesses the group's strength
and judges a fight too costly to risk.



Conflict between groups falls
war-avoidance fraction $r > 0$

LAYER 3 Cooperation

Partners see a reliable group;
exchange and alliance ties form.



Cooperation networks buffer crises
survival differential $\alpha_{\text{eff}} < \beta_{\text{eff}}$

Reinforcement
*Layer 2 and 3 benefits raise
group fitness, sustaining
the practice*

**Environmental
uncertainty**
*raises the value
of deterrence
and cooperation*

Monuments arise from individual signaling within groups, not from coercion or group-level selection.

Within-group peer rewards drive construction; between-group audiences add deterrence and exchange benefits.